

## GRADE 6

- 1) Simplify  $5a \times (-3a)$ .

Sarina kept a record of her after – school earnings. (Question 2-3)

| Number of weeks worked | 1    | 2    | 3    | 4    |
|------------------------|------|------|------|------|
| Total Saved            | £ 14 | £ 28 | £ 42 | £ 56 |

- 2) Use logical reasoning and arithmetic to find how much Sarina will have saved after working for 5 weeks.
- 3) Use logical reasoning and arithmetic to find how many weeks Sarina will have to work to save £ 98.
- 4) The Wolverines swimming team won a total of 15 first – and second place medals at their last swim meet. If they won 7 more first – place medals than second – place medals, how many first – place medal did they win?

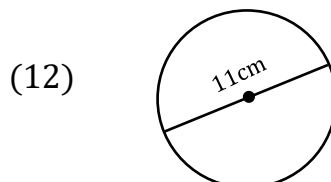
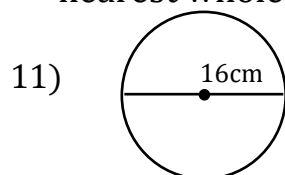
Evaluate each expression for  $a = 72$ ,  $b = 28$  and  $c = 8$  (Question 5-6)

- 5)  $b \times 7$
- 6)  $a \times b \times c$ .
- 7) Tamara, David, Elena and James entered their dogs in the country dog show. The dogs were terrier, a setter, and a golden retriever, and a Great Dane. Neither boy entered a setter. Tamara owns a golden retriever. What breed of dog did Elena enter in the show? (Solve by using logical reasoning).

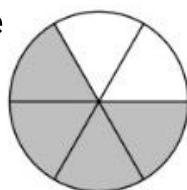
Solve each equation using mental math.

- 8)  $n \times 8 = 56$
- 9)  $h \div 20 = 35$
- 10) Simplify the expression  $2x + x^2 - 7 - 8x$ , then evaluate for  $x = 5$ .

Find the circumference of the circles. Use 3.14 or  $\frac{22}{7}$  for  $\pi$ . Round to the nearest whole number.



For question 13 – 14 use this figure



13. Find the ratio of unshaded sections to shaded sections

14. Find the ratio of shaded sections to all the sections.

Write an algebraic expression for the word expression (question 15 – 17)

15. 47 less than the product of  $y$  and 7

16. The difference between  $a$  and 4 divided by the sum of  $b$  and 9.

17.  $K$  times 12 divided by the product of  $d$  and 3.

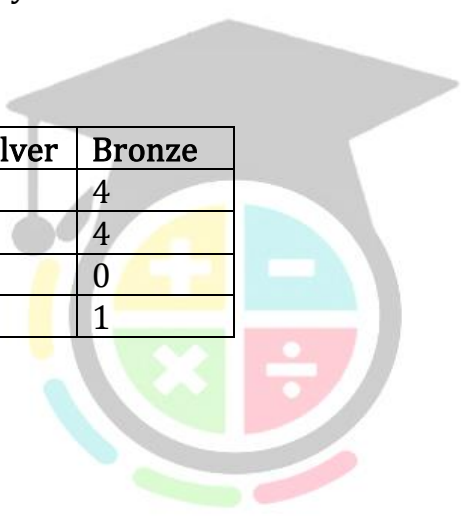
Use mental math to find the value of  $y$  (question 18 and 19).

18.  $Y \times -2 = -16$

19.  $-5 \times y = 30$ .

20. Draw a graph to display the data in this table.

Olympics Medals won



| Athletics   | Gold | Silver | Bronze |
|-------------|------|--------|--------|
| L. Latynina | 9    | 5      | 4      |
| T. Ono      | 5    | 4      | 4      |
| P. Nurmi    | 9    | 3      | 0      |
| S. Kato     | 8    | 3      | 1      |